

Solution Architecture

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Layer	Components	Responsibility
Presentation	HTML/CSS/JavaScript + Jinja2	Collect user input and display results.
API	FastAPI + Uvicorn	Validate requests, expose endpoints, route operations.
Feature Modules	Q&A, Explain, Summarize, Quiz, Learning Path	Construct task-specific prompts and structure outputs.
AI Service	Gemini Service Layer	Discover models, select primary/fallback model, call Gemini.
Fallback	Offline/Demo Explainer	Provide simplified response when live API is unavailable.
Configuration	python-dotenv / .env	Supply secrets and runtime settings securely.

1) Student submits request → 2) FastAPI validates → 3) Feature module builds prompt → 4) Gemini Service Layer selects model → 5) Gemini generates response → 6) Feature module structures result → 7) Frontend displays result. Failure of live AI triggers model fallback or offline/demo behavior.